

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			As the water is heated {it expands / its volume increases} and density decreases (1) The cooling water's volume decreases and its density increases (1) Or density decreases when [water] heated and density increases when it cools(1).. .. because {it / water} expands when heated and contracts when cools (1) Not: <u>particles</u> expand or become less dense	2			2		2
	(b)			Indicative content: Ion/lattice process 1. the molecules/ions at the “high temperature end” of the substance gain energy 2. and vibrate with increasing amplitude / kinetic energy. 3. They transfer their energy to other neighbouring molecules/ions by collisions and this continues through the body of the material. Free-electron process 4. Copper has free (or de-localised or mobile) electrons which gain energy at the “high temperature end” 5. and move / migrate towards the “lower temperature end”, 6. transferring energy more quickly 5 – 6 marks Expect at least 4 statements with at least two from each process A correct, description of the two processes that identify metals as good conductors of thermal energy is given. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		6